

Ordinary Differential Equations

Introduction to Differential Equations

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ENGR 213**

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Today's Lecture

- What are differential equations and Why we use them
- Definitions and Terminology

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Section 1.1**

Differential Equations

Why do we learn differential equations?

- ❑ The world does not just have values, it changes.

Physical system → Laws governing change → Prediction

e.g.: A hot object → heat leaves the object → object temperature
in time

- ❑ *Differential equations* describe how quantities change.
- ❑ Once we know the rules governing change, we can predict, design, and control engineering systems.

Differential equation = A mathematical model of change

Where do differential equations appear?

- Cooling a hot object

How fast is the temperature changing? $\rho cV \frac{dT}{dt} = -hA(T - T_{\infty})$

- Vehicle Motion

How fast is the velocity changing? $\frac{dx}{dt} = v \quad \frac{dv}{dt} = a$

- Chemical Reaction ($A \rightleftharpoons B$)

How fast is concentration of A changing? $\frac{dC_A}{dt} = -k_1 C_A + K_2 C_B$

- Population growth

How does a population change when resources are limited? $\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$

Different physical systems, different equations → But the common feature is: an unknown quantity is related to one or more of its derivatives.



Definitions and Terminology

- A **differential equation (DE)** is an equation containing the derivatives of one or more *dependent variables* with respect to one or more *independent variables*.

e.g.:
$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

- P is the dependent variable
- t is the independent variable
- $\frac{dP}{dt}$ is the derivative of y with respect to t

How do we solve such an equation for the unknown function $P = \phi(t)$?

- DEs can be classified by:

- Type
- Order
- Linearity

□ Classification by Type

- **Ordinary differential equation (ODE):** Derivatives are with respect to a **single independent variable**.

e.g.,: $\frac{dx}{dt} + \frac{dy}{dt} = 3x + 2y$

$$\frac{dy}{dx} + y = x$$

$$\frac{dy}{dx} + 6y = e^{-x}$$

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 0$$

- **Partial differential equation (PDE):** Derivatives are with respect to **two or more independent variables**.

e.g.,: $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - \frac{\partial u}{\partial t}$$

- **Notation:** Throughout this course, ordinary derivatives will be written using either the *Leibniz notation* dy/dx , d^2y/dx^2 , d^3y/dx^3 , ..., d^ny/dx^n or the *prime notation* y' , y'' , y''' , $y^{(4)}$, ..., $y^{(n)}$.

$$\text{e.g.} \therefore \frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 0 \longleftrightarrow y'' + y' - 12y = 0$$

- *Newton's dot notation* is sometimes used to denote derivatives with respect to time t . e.g. $\dot{y}$

- Partial derivatives are often denoted by a *subscript notation* indicating the independent variables.

$$\text{e.g.} \therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longleftrightarrow u_{xx} + u_{yy} = 0$$

□ Classification by Order

- The order of a differential equation (ODE or PDE) is the order of the **highest derivative** in the equation.

$$\text{e.g.,: } \frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 4y = e^x \qquad 2 \frac{\partial^4 u}{\partial x^4} + \frac{\partial^2 u}{\partial t^2} = 0$$

□ General form vs. Normal form for ODEs:

$$F(x, y, y', \dots, y^{(n)}) = 0 \longrightarrow \frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$$

(F is a real-valued function, and the equation is in general form)

(f is a real-valued continuous function, and the equation is referred to as **normal form**)

$$\text{e.g.,: } (y - x)dx + 4xdy = 0 \longrightarrow \frac{dy}{dx} = \frac{x - y}{4x}$$

$$\text{e.g.,: } (y^2 + x^2)dx + 6xydy = 0 \longrightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{-6xy}$$

□ Classification by Linearity

- An n th-order ODE, $F(x, y, y', \dots, y^{(n)}) = 0$, is linear in the variable y if F is linear in $y, y', \dots, y^{(n)}$.

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

➤ Characteristics of a linear ODE:

- I. The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the first degree (power of each term involving y is 1.)
- II. The coefficients $a_0, a_1, \dots, a_n$ of $y, y', \dots, y^{(n)}$ depend at most on the independent variable x .
- III. Nonlinear functions of the dependent variable or its derivatives such as $\sin(y)$ or e^y cannot appear in a linear equation.

e.g.: $(y - x)dx + 4xdy = 0$

$$\frac{d^2y}{dx^2} + \sin(y) = 0$$

$$(y - 1)y' + 4y = 2x^2$$

$$y'' - 2y' + y = 0$$

$$x^3 \frac{d^3y}{dx^3} + 3x \frac{dy}{dx} - 5y = e^x$$

$$\frac{d^4y}{dx^4} + y^2 = 0$$



□ Solution of an ODE

- Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a solution of the equation on the interval.

$$F(x, y, y', \dots, y^{(n)}) = 0$$



function $\phi(x)$ is a solution
of this ODE if

- I. $\phi(x)$ possesses at least n derivatives
 - II. $F(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)) = 0$ (for all x in I)
- The interval I is also called the interval of definition or the domain of the solution and can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

e.g.: $y'' + y = 0$

$\phi(x) = \sin(x)$  proposed solution

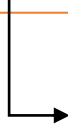
I. $\phi(x)$ possesses at least n derivatives:

✓ $\phi(x)$ has 2 derivatives: $\phi'(x) = \cos(x)$ $\phi''(x) = -\sin(x)$

II. $F(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)) = 0$ (for all x in I)

✓ Substitute into the ODE: $\phi''(x) + \phi(x) = -\sin(x) + \sin(x) = 0$

$\therefore \phi(x) = \sin(x)$ is a solution on interval $I = (-\infty, \infty)$

 In I , $\phi(x)$ is defined, and the derivatives of $\phi(x)$ exist and are continuous

$\phi(x) = 0$ is also a trivial solution on interval I

 A solution of the DE that is identically zero on the interval

e.g.: $y'' = \frac{2}{x^3}$

$\phi(x) = \frac{1}{x}$  proposed solution

I. $\phi(x)$ possesses at least n derivatives:

✓ $\phi(x)$ has 2 derivatives: $\phi'(x) = \frac{-1}{x^2}$ $\phi''(x) = \frac{2}{x^3}$

II. $F(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)) = 0$

✓ Substitute into the ODE: $\phi''(x) = \frac{2}{x^3}$

But at $x=0$: $\phi(0) = \frac{1}{0}$ is undefined.

$\therefore \phi(x) = \frac{1}{x}$ is a solution on the interval $(-\infty, 0)$ or $(0, \infty)$

□ Explicit and Implicit Solutions

- A solution in which the dependent variable is expressed solely in terms of the independent variable and constants is said to be an **explicit solution**.

e.g.: $y'' - 2y' + y = 0 \longrightarrow y(x) = \sin(x)$ explicit solution

- A relation $G(x, y) = 0$ is said to be an **implicit solution** of an ODE on an interval I provided there exists at least one function ϕ that satisfies the relation as well as the differential equation on I .

e.g.: $\frac{dy}{dx} = -\frac{x}{y} \longrightarrow x^2 + y^2 = 25$ implicit solution



□ Families of Solutions

- Like integral calculus, we usually obtain a solution to a first-order ODE containing an arbitrary constant c .

$$\text{Integral: } \int 2x dx = x^2 + C \quad \xrightarrow{\text{Why is there a } C?} \quad \frac{d}{dx}(x^2 + C) = 2x$$

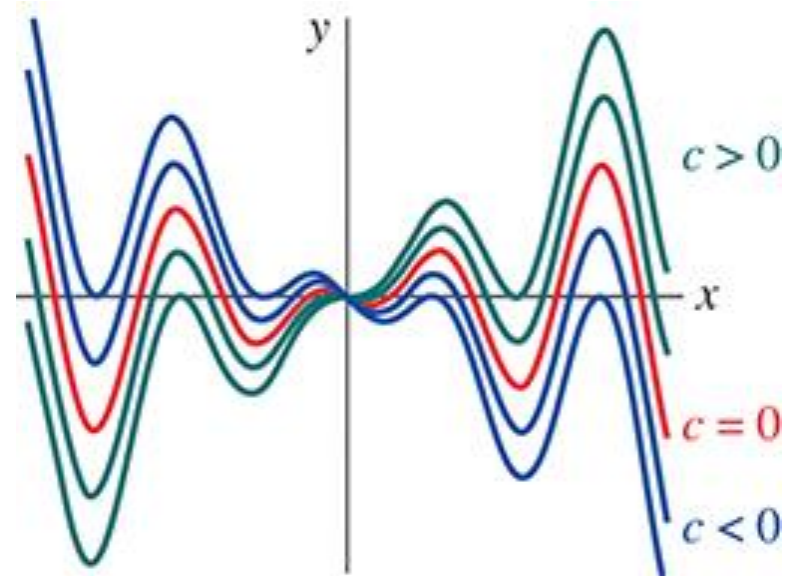
$$\text{ODE: } \frac{dy}{dx} = 2x \quad \text{solution: } y = (x^2 + C) \quad * \text{ so, this ODE has a family of solutions, one for every value of } C$$

- A solution with a constant C represents a set $G(x, y, c) = 0$ of solutions called a **one-parameter family of solutions**.
- When solving an n th-order DE $F(x, y, y', \dots, y^{(n)}) = 0$, we seek an **n -parameter family of solutions** $G(x, y, c_1, c_2, \dots, c_n) = 0$.

$$\text{ODE: } xy' - y = x^2 \sin(x)$$

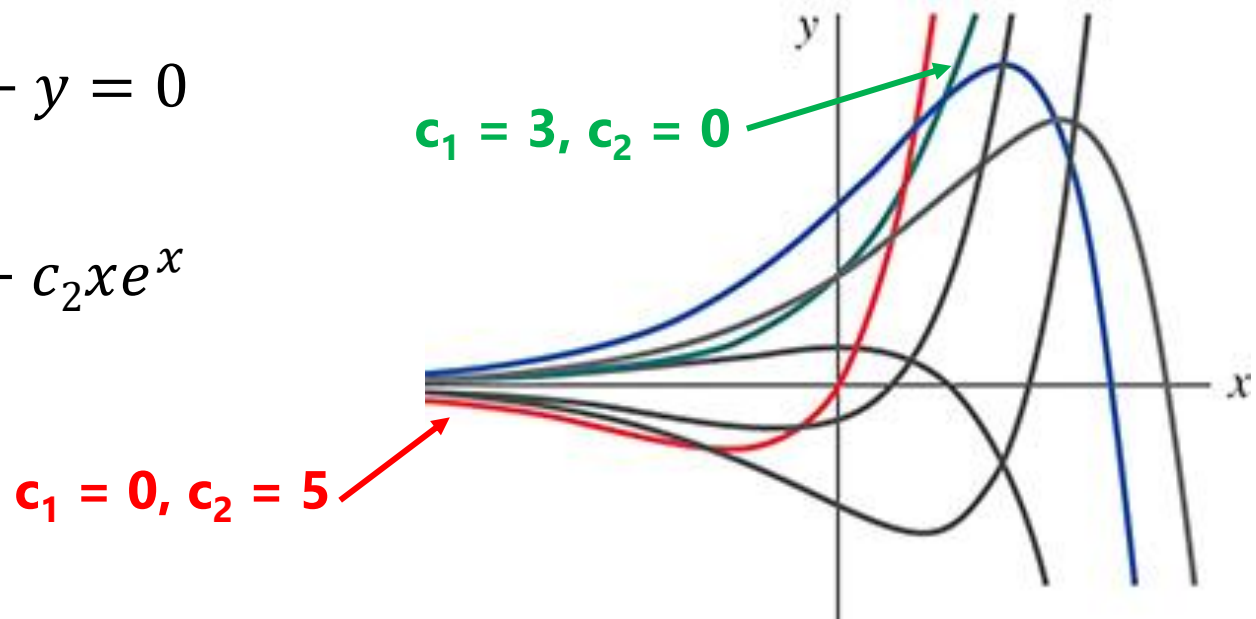
$$\text{Solution: } y = cx - x \cos(x)$$

- ✓ The graphs of some particular solutions in this family for various choices of c .



$$\text{ODE: } y'' - 2y' + y = 0$$

$$\text{Solution: } y = c_1 e^x + c_2 x e^x$$



□ Systems of Differential Equations

- A system of ODEs is two or more equations involving the derivatives of two or more unknown functions of a single independent variable.

$$\frac{dx}{dt} = f(t, x, y) \quad \frac{dy}{dt} = g(t, x, y)$$

- A solution of a system is a set of functions defined on a common interval I that satisfies each equation of the system on this interval.

e.g.:

system of ODEs $\left\{ \begin{array}{l} \frac{dx}{dt} = x + 3y \\ \frac{dy}{dt} = 5x + 3y \end{array} \right.$ $\longrightarrow$ solutions $\left\{ \begin{array}{l} x = e^{-2t} + 3e^{6t} \\ y = -e^{-2t} + 5e^{6t} \end{array} \right.$